

File permissions in Linux

Project description

In this project, I act as a security professional responsible for managing file permissions in a Linux environment for a research team. My goal was to ensure that each file and directory had the correct access levels along with removing any unauthorized access and assigning appropriate permissions to team members. I used common Linux commands like `ls -la`, `chmod`, `cd`, etc. to inspect and correct permissions for all three levels of control User, Group, and Other.

Check file and directory details

To check file and directory details first navigate to the specific subdirectory that you need to display the permissions for and what is within the directory using `cd`. You can use the `ls` command to see all the files within the given directory. Using `ls -la` will display all the permissions for each file even those that are hidden.

Since we are assigning permissions to those users in the projects file within the working directory, we need to first navigate to the projects subdirectory. To get here we type

```
cd /home/researcher2/projects
```

Now that we've navigated to the project's directory, we need to display all the current permissions by using the command `ls-l` or `ls-la` (For hidden files)

```
drwxr-xr-x 3 researcher2 research_team 4096 Jul  3 13:48 .
```

```
drwxr-xr-x 3 researcher2 research_team 4096 Jul  3 14:47 ..
```

```
-rw--w---- 1 researcher2 research_team  46 Jul  3 13:48 .project_x.txt
```

```
drwx--x--- 2 researcher2 research_team 4096 Jul  3 13:48 drafts
```

```
-rw-rw-rw- 1 researcher2 research_team  46 Jul  3 13:48 project_k.txt
```

```
-rw-r----- 1 researcher2 research_team  46 Jul  3 13:48 project_m.txt
```

```
-rw-rw-r-- 1 researcher2 research_team  46 Jul  3 13:48 project_r.txt
```

```
-rw-rw-r-- 1 researcher2 research_team  46 Jul  3 13:48 project_t.txt
```

Describe the permissions string

The permissions string is a 10-character sequence of letters that explains first if it is a normal directory with the letter `d` or `-` for a file - then has letters signifying whether the user, group, or other has permissions to read, write and execute files.

It looks like this: `drwxr-xr-x`

It starts with the file type (`d` for directory, `-` for regular file). The next three characters represent the permissions for the **user** (owner) of the file. The subsequent three characters are for the **group** that owns the file. Finally, the last three characters represent permissions for **others** (everyone else). Within each set of three characters:

r indicates read permission.

w indicates write permission.

x indicates execute permission.

- indicates no permission.

Change file permissions

In order to change file permissions you first have to display them as mentioned previously with either `ls-l` or `ls-la` (For hidden files). Once you can see the 10 character string and have noted what each the User, Group, and Other have permissions to already we can use the command `chmod`. In order to negate a specific command, we use the `-` and to add we use plus. An example if we were adding reading rights to the User would look like `chmod u+r`. You can also stack multiple changes for file permissions by placing a comma in between them. For example, `chmod u+r, g-w, o-r (File Name)`

Change file permissions on a hidden file

It is a very similar process except when displaying the files we must use the commands `ls -la` to show all files including the hidden ones. The like normal we'd change the permissions using `chmod`.

Change directory permissions

Organizations only want the researcher2 user to have access and permissions in the draft's directory. It is similar to changing the permission of a file except instead you put the directory name at the end instead. For example, `chmod g-x drafts` will remove the executable permissions so only researcher2 has them.

Summary

I changed multiple permissions to match the level of authorization my organization wanted for Files and directories in the project's directory. The

The first step in this was using `ls -la` to check the permissions for the directory as well as show the hidden files. I then used the `chmod` command multiple times to change the permissions on files and directories.